Max Lasek

631-764-1264 | Centereach, NY | maxlasekeng@gmail.com | linkedin.com/in/max-lasek | Portfolio

Education

University at Buffalo, State University of New York Expected May 2029
B.S. Mechanical Engineering | GPA: 3.2 / 4.0

Engineering Projects

Robotic End Effector | *UB Space Bulls (URC)* Aug. 2025 – Present
- Raised clamping torque for 5 kg lifts by replacing a lead-screw concept with a worm-driven 4-bar, using a 24:1 reduction from a 25 kg·cm HiWonder servo for a 2.5× clamping torque safety margin
- Reduced heavy friction and gear backlash found in testing by iterating gear designs and tolerances and adding brass heat-set inserts and ball bearings
- Designing TPU grips over the PETG fingers to increase the grip factor of safety; researching grip styles, including concave pads and compliant adaptive grips
- Simplified assembly by standardizing all fasteners on M4, and designed the mount to integrate with the team's differential wrist

3-DOF Practice Arm | *UB Space Bulls (URC)* Apr. 2026 – May 2026
- Scoped a 5-DOF replica down to its 3 major joints and defined specifications up front: 12 / 9 / 2 in link lengths, 26.9 / 6.5 / 0.2 lb·in torque at the shoulder, elbow, and wrist, and a DH table for kinematics
- Reduced structural weight by running FEA on all major parts and removing material where stress was low; switched to dual beams per side to increase stiffness
- Eliminated singularities in the joint layout to reduce points of failure, so the software team could reliably test joint control
- Assembled the main body and helped assemble the gearboxes; wired motors, motor controllers, and Arduinos

RC / Autonomous Vehicle | *Personal Project* Jun. 2026 – Present
- Diagnosing distance overshoot in a cascaded PID controller (waypoint steering and wheel speed) using encoder odometry in Python on a Raspberry Pi 5
- Ran FEA on multiple TPU wheel designs, simulating deformation under load to compare the flexibility of different shapes
- Rearranged electronics into a compact, weight-balanced layout, mounting the 2S LiPo low for a low center of mass, with a buck converter regulating 5V to the Pi
- Chose a Pi over an Arduino to allow Bluetooth and camera expansion; replaced servo steering with swivel casters to meet parts and schedule constraints

Experience

Manipulation Team Lead | *University Rover Challenge, UB Space Bulls* Sep. 2025 – Present
- Lead a 15+ member arm team designing and manufacturing a 5-DOF robotic arm, working across its mechanical, electrical, and software systems
- Run design reviews and facilitate integration between sub-teams, including the rail system attaching the arm to the chassis, electrical systems, and the interfaces between each joint
- Rebuilt onboarding over the summer and trained 6+ new members in CAD, fabrication, wiring, research, team software, and prototyping, creating a structure that helps them stay and contribute

Project Lead | *VEX Robotics, Centereach High School* Sep. 2021 – May 2025
- Led teammates through machining, construction, and field setup; rebuilt robots each season around new challenge requirements using motors, sensors, and pneumatics
- Added a rubber band suspension to the encoder wheels so they stayed on the ground, keeping the robot's position tracking accurate for autonomous movements

Technical Skills

Languages/Tools: Python, C++, Java, MATLAB
Mechanical: CAD, FEA, GD&T, DFM, tolerance analysis, gear train design, kinematics (DH), 3D printing, machining, basic CNC/CAM, laser cutting
Electrical: Arduino, Raspberry Pi 5, PID control, motor control, soldering, power distribution, voltage regulation
Applications: Fusion 360, SolidWorks, Prusa Slicer
Leadership: Eagle Scout